

Data Structures in Python

- ↳ Linear data structure
 - ↳ Non-linear data structure
- a way to store the data in an efficient & structured manner

List & String → Linear Data Structure

Python internal working

name = "Priya"

↳ String → Composed of multiple characters

name →

P	r	<u>i</u>	y	a
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0 1 2 3 4

-5 -4 -3 -2 -1 → negative indexing

name[2] = i

name[-1] = a

name[4] = a

h	e	l	l	o		w	o	r	l	d
0	1	2	3	4	5	6	7	8	9	10
-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1

dlrow olleh

hlowrd → Output

$$\left\{ \begin{array}{l} -1-1 = -2 \\ -2-1 = -3 \\ -3-1 = -4 \\ -4-1 = -5 \end{array} \right.$$

$$\left. \begin{array}{l} \text{start} = 0 \\ \text{end} = 11 \\ \text{step} = 2 \end{array} \right\}$$

Mutability → can be changed

Immutability → cannot be
changed

List	0	1	2	3	4	5
	10	23.5	"Priya"	101	253	33.5
	-6	-5	-4	-3	-2	-1

indexing

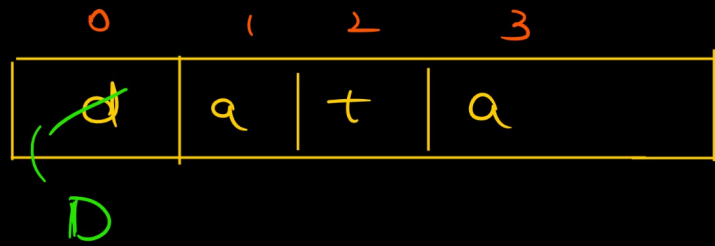
Positive

Negative

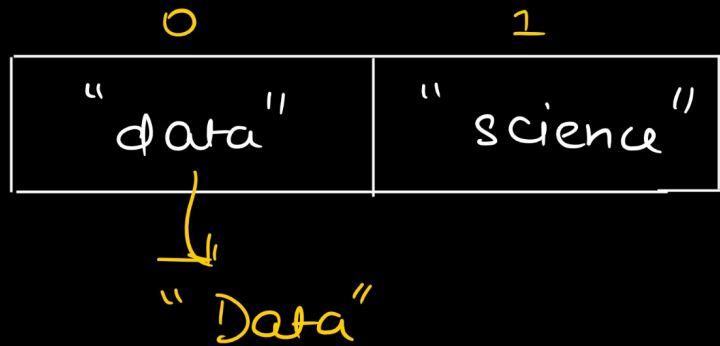
slicing operation

String $\rightarrow$ immutable

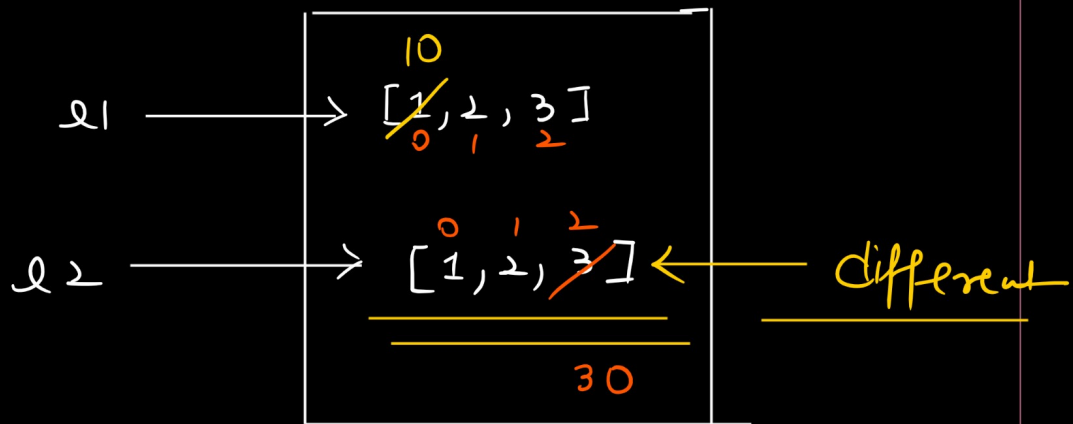
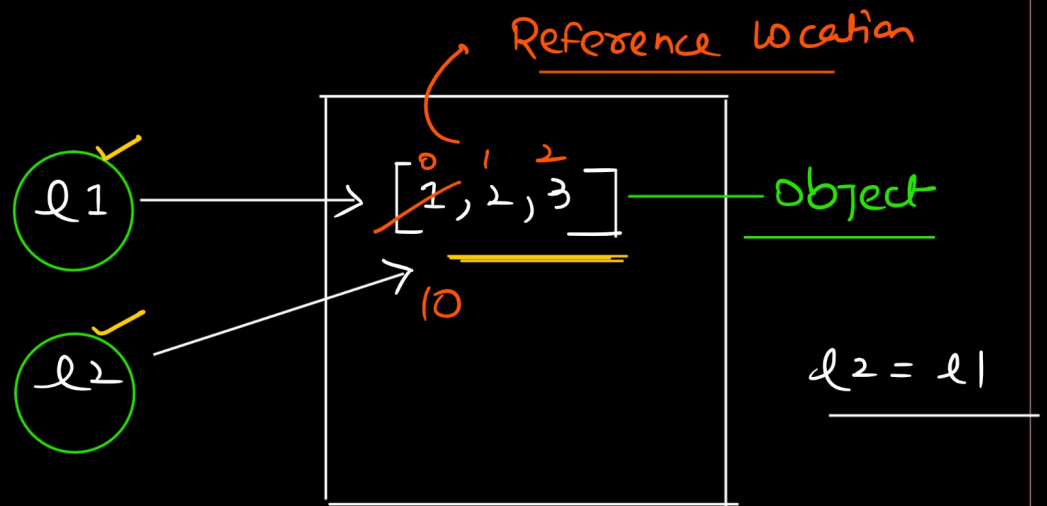
you cannot change after creation



List $\rightarrow$ mutable



Python internal working



String1 → a p p l e

String2 → b a n a n a

$String1 < String2$

↳ true

$a < b$ → ASCII code

97 < 98 Value

string_3.index('world')

3

hi world

0	1	2	3	4	5	6	7
h	i		w	o	r	l	d

